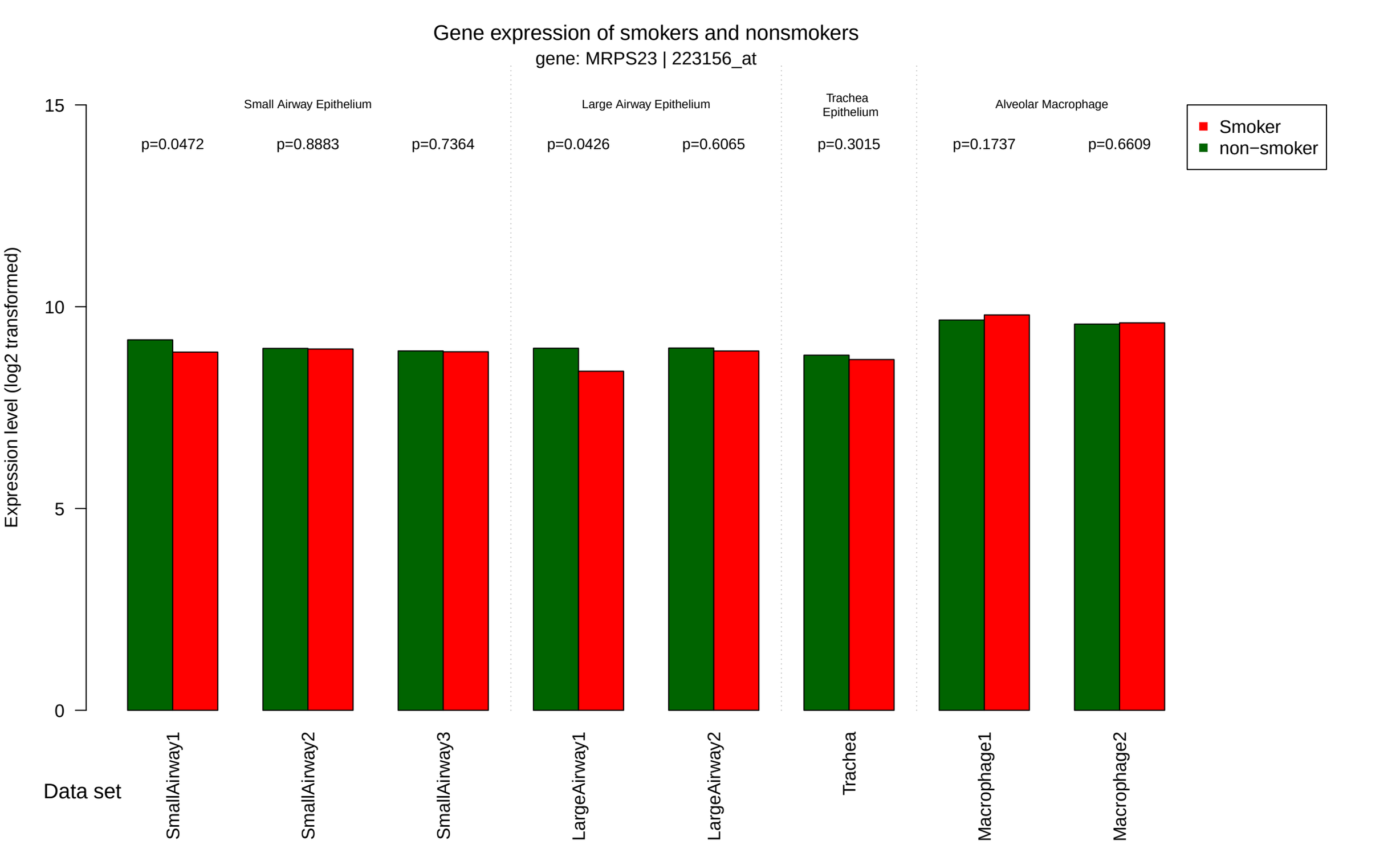
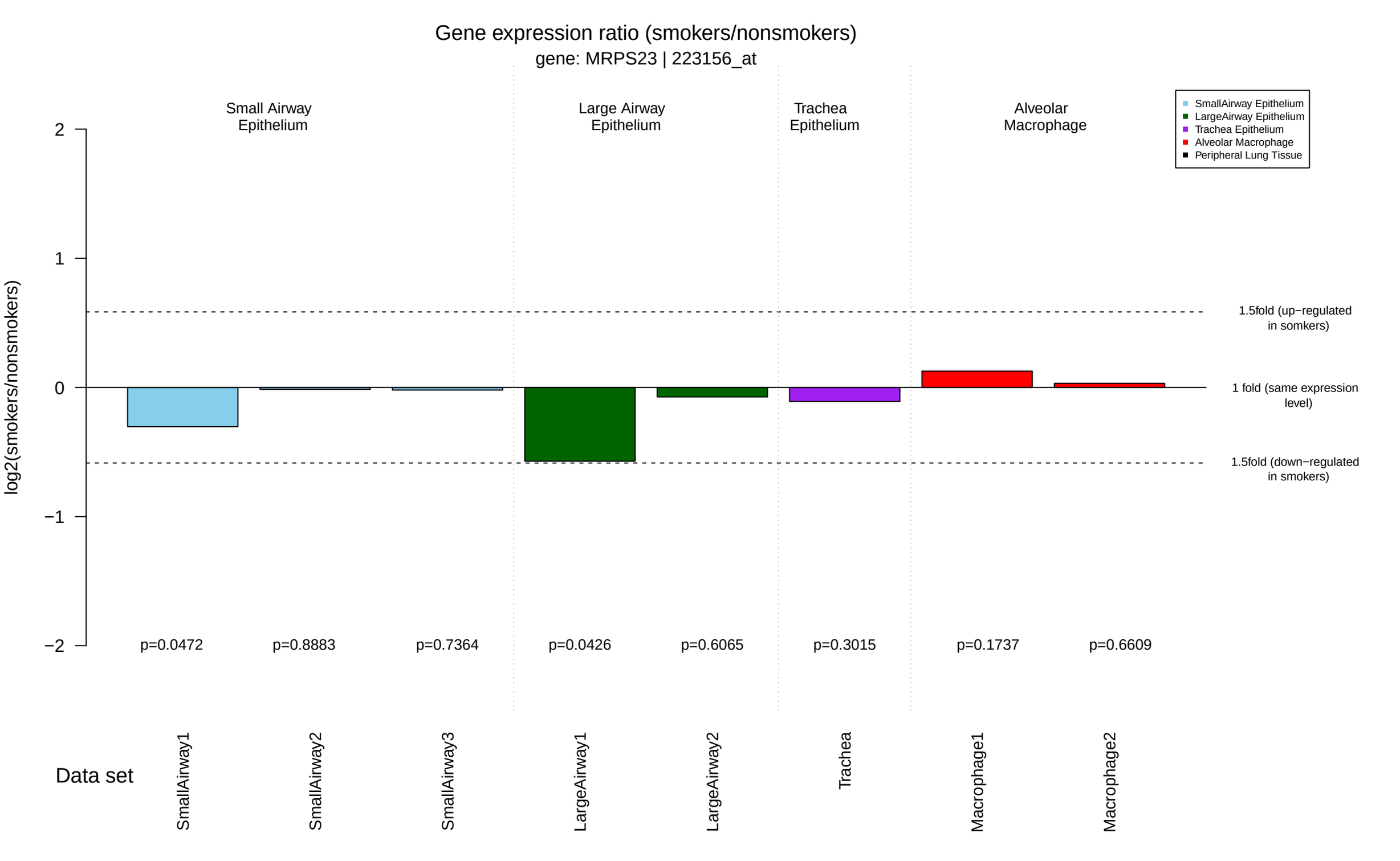
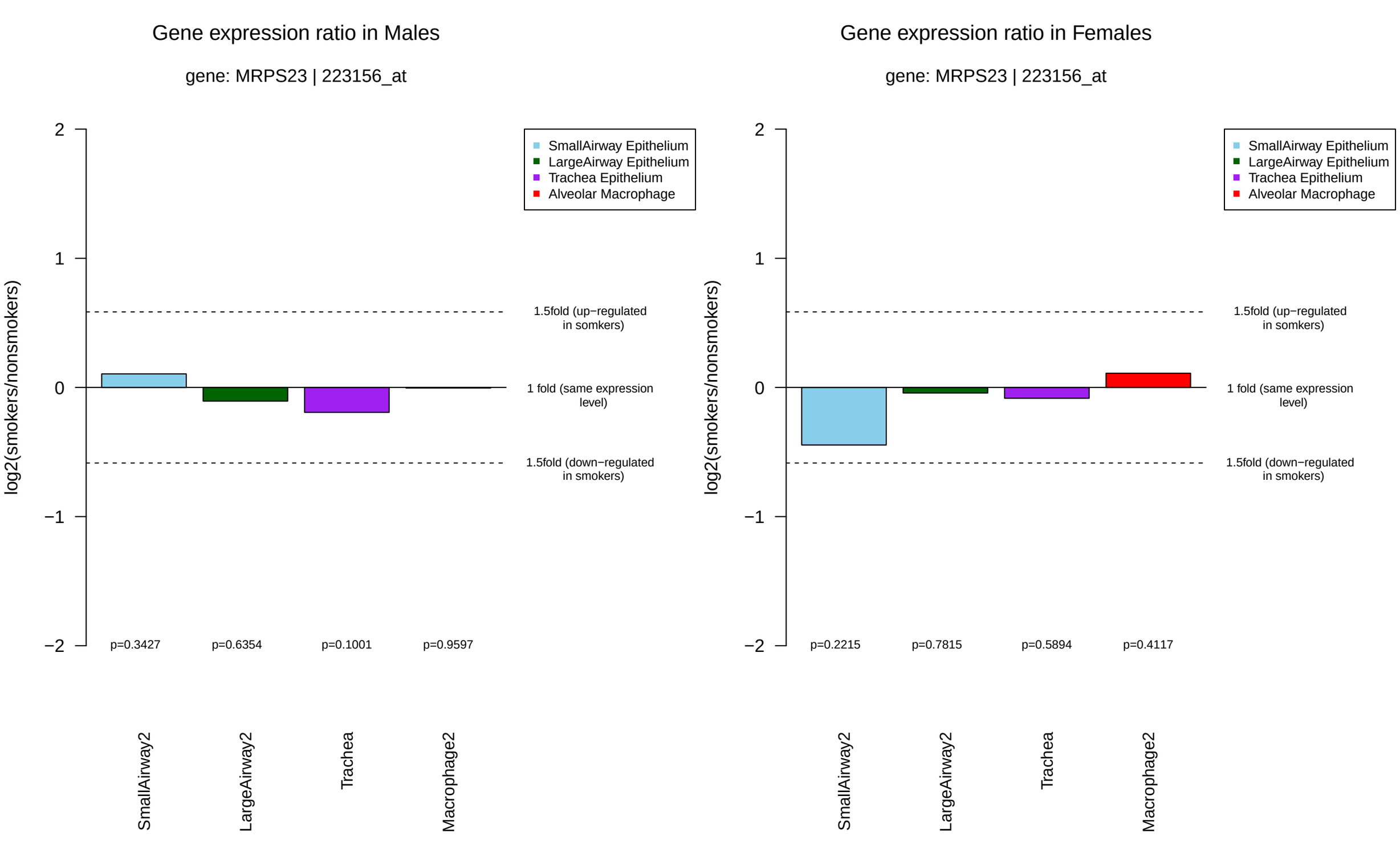
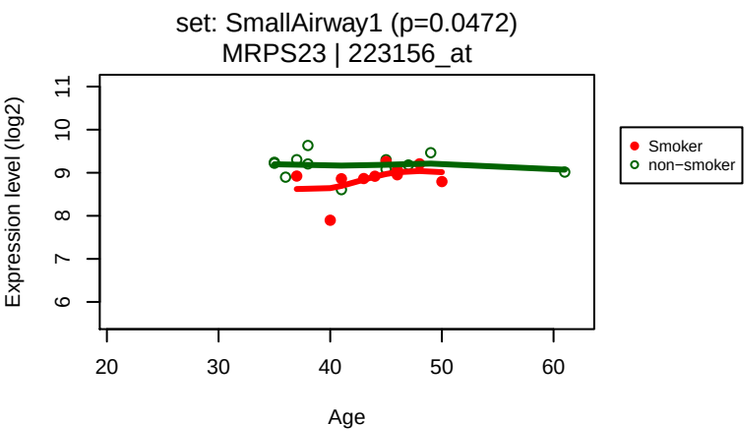
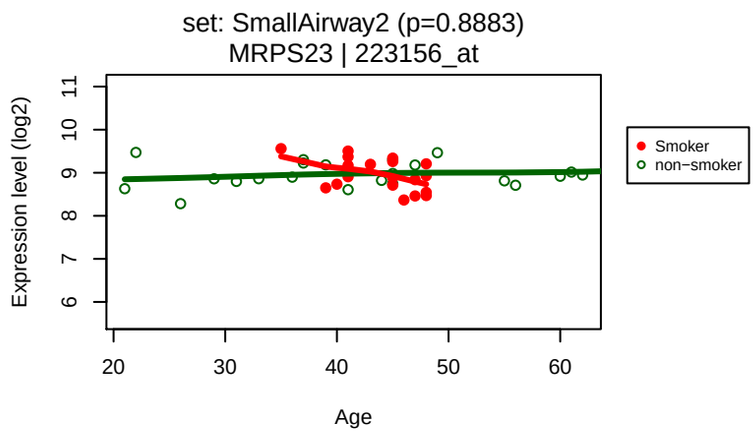
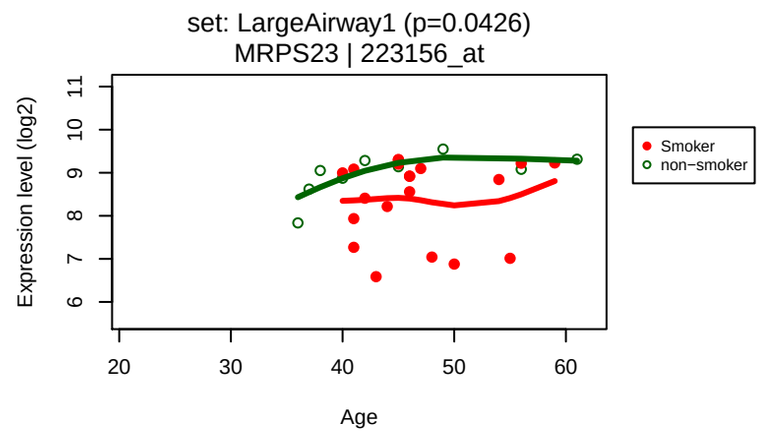

Gene expression of smokers and nonsmokers
gene: MRPS23 | 223156_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0472
p=0.8883
p=0.7364
p=0.0426
p=0.6065
p=0.3015
p=0.1737
p=0.6609
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

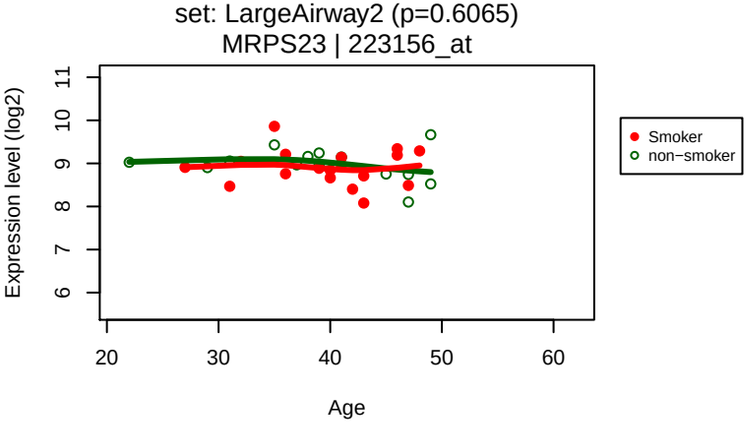

Gene expression ratio (smokers/nonsmokers)
gene: MRPS23 | 223156_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0472
p=0.8883
p=0.7364
p=0.0426
p=0.6065
p=0.3015
p=0.1737
p=0.6609
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

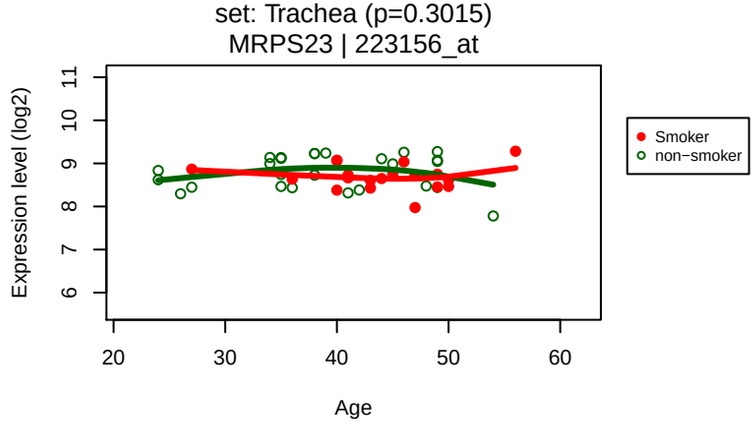

Gene expression ratio in Males
gene: MRPS23 | 223156_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3427
p=0.6354
p=0.1001
p=0.9597
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPS23 | 223156_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2215
p=0.7815
p=0.5894
p=0.4117
SmallAirway2
LargeAirway2
Trachea
Macrophage2

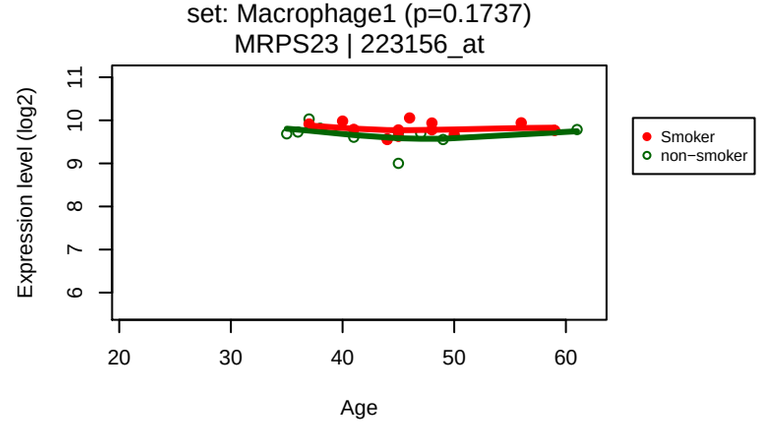

set: SmallAirway1 (p=0.0472)
MRPS23 | 223156_at
Expression level (log2)
Age
Smoker
non−smoker

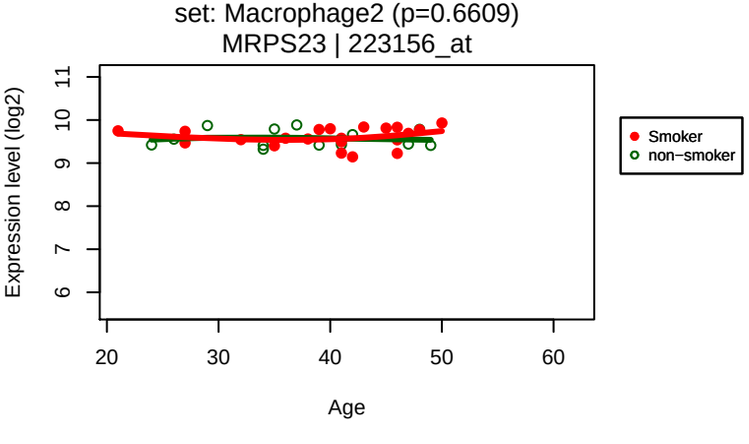

set: SmallAirway2 (p=0.8883)
MRPS23 | 223156_at
Expression level (log2)
Age
Smoker
non−smoker

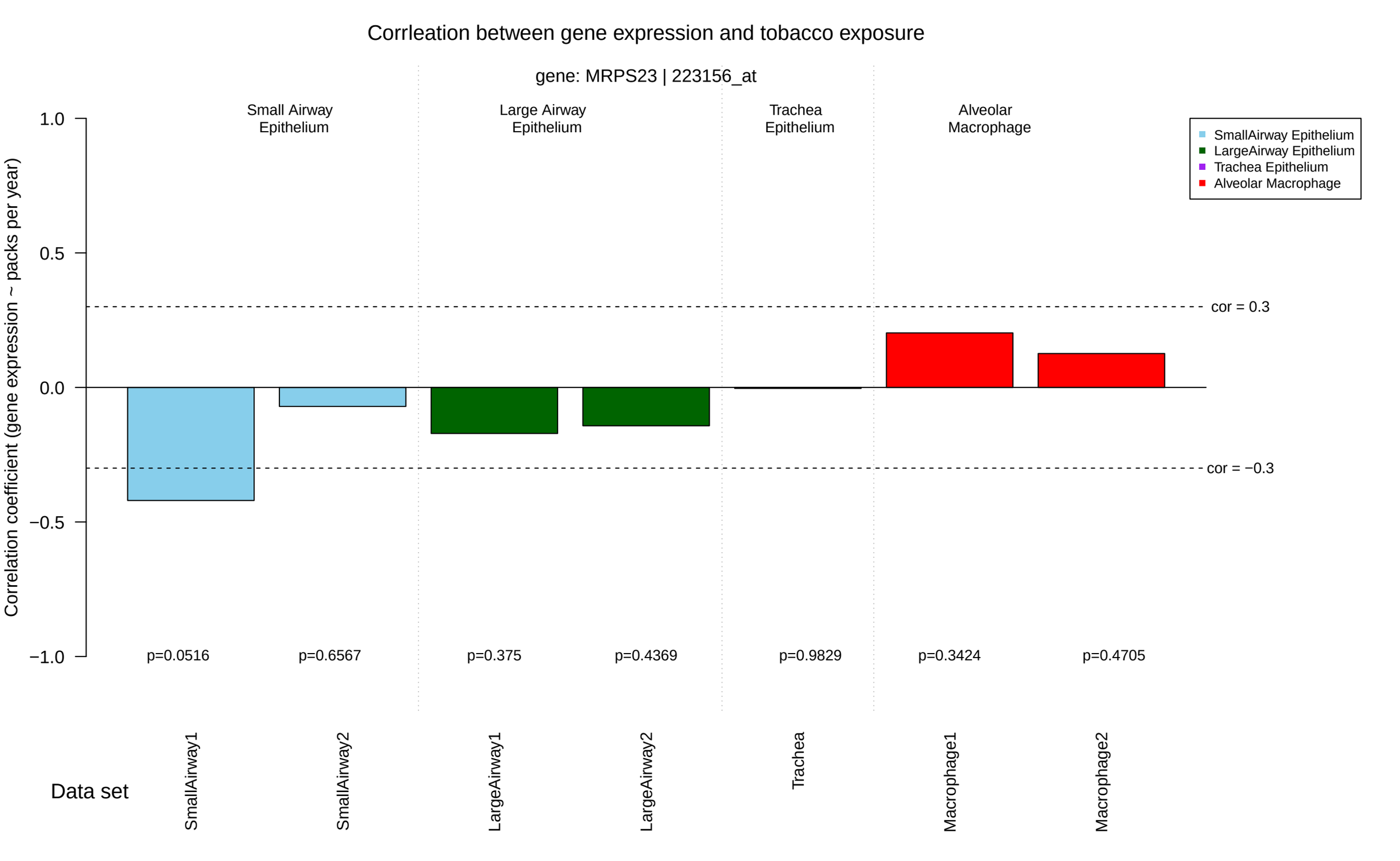

set: LargeAirway1 (p=0.0426)
MRPS23 | 223156_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6065)
MRPS23 | 223156_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3015)
MRPS23 | 223156_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1737)
MRPS23 | 223156_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6609)
MRPS23 | 223156_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPS23 | 223156_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0516
p=0.6567
p=0.375
p=0.4369
p=0.9829
p=0.3424
p=0.4705
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2